CE205 Databases and Information Retrieval

Weighting and Evaluation of Information Retrieval System

Class 8 (Week 9)

Objective

The purpose of this exercise is to experiment with TF and IDF scores in a worked example to understand how they work in a practical case. In addition, how precision and recall affect each other.

Problem 1:

Given a document D containing terms with given frequencies:

A(3), B(2), C(1)

Assume the collection contains 10,000 documents and document frequencies of these terms are:

A(50), B(1300), C(250)

Determine tf-idf weight for the terms A, B and C using the following formula.

$$tf_idf_{t,d} = tf_{t,d} \times log_{10}\left(\frac{N}{df_t}\right)$$

Solution:

Term	Document Frequency (df_t)
A	50
B	1300
C	250

Term Frequencies ($tf_{t,d}$)	
Term	Document D
A	3
B	2
C	1

Compute *tf-idf* using the given formula

tf-idf	
Term	Document D
A	
B	
C	

Problem 2

Look at the two passages below; they are your 'documents'; the two documents taken together constitute your 'collection'.

For each passage take the words 'Scarlatti' and 'sonatas' and compute tf, idf and hence tf-idf. What differences in the values of tf, idf and tf-idf do you see between the passages for the two terms?

Example Document 1

Three centuries ago a man named Domenico ***Scarlatti*** *churned out an enormous number of keyboard* ***sonatas*** *— more than 550. In his time, only a fraction of his* ***sonatas*** *were known .*

Example Document 2

Five hundred and fifty five keyboard ***sonatas*** *have secured the reputation of Domenico* ***Scarlatti*** *. Despite their huge number, Scarlatti's* ***sonatas*** *are always full of surprises. Harpsichordist Ralph Kirkpatrick, who lent his initial to the 'K' numbers by which Scarlatti's works are catalogued, has observed that none of the* ***sonatas*** *is typical .*

Solve the problem using non-normalized as well as logarithmic tf-idf.

Solution:

Term	Document Frequency (df_t)
Scarlatti	
sonatas	

Term Frequencies ($tf_{t,d}$)		
Term	Document 1	Document 2
Scarlatti		
sonatas		

The formula for non-normalised *tf-idf* is

$$tf_idf = tf_{t,d} \times \frac{1}{df_t}$$

tf-idf		
Term	Document 1	Document 2
Scarlatti		
sonatas		

The formula for logarithmic *tf-idf* is

$$tf_idf = [1 + \log(tf_{t,d})] \times \log\left(\frac{N}{df_t}\right)$$

tf-idf		
Term	Document 1	Document 2
Scarlatti		
sonatas		

Problem 3

Find Precision and Recall in both cases. Discuss the findings, and can you draw any conclusions?

Case1:

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 80 (True Positive)**	**FP = 800 (False Positive)**
	Not Relevant	**FN = 0 (False Negative)**	**TN = 999,240 (True negative)**

$$Precision = \frac{TP}{TP + FP} =$$

$$Recall = \frac{TP}{TP + FN} =$$

Case2:

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 5 (True Positive)**	**FP = 10 (False Positive)**
	Not Relevant	**FN = 75 (False Negative)**	**TN = 1,000,030 (True negative)**

$$Precision = \frac{TP}{TP + FP} =$$

$$Recall = \frac{TP}{TP + FN} =$$

Problem 4

The tf-idf weights for the terms in various documents are given as follows:

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

What will be the score of each document for the following queries?

i) Why does Calpurnia fear a worser fate for Caesar?

ii) How does Antony describe mercy in contrast to Brutus?

What will be the order of the retrieved documents for each query?

$$\text{Score}(q,d) = \sum_{t \in q \cap d} \text{tf.idf}_{t,d}$$